8 The nuclide notation for the radioactive isotope carbon-14 is $^{14}_{6}C$.

(a) Using the symbols shown in Fig. 8.1, draw a diagram to show the number of electrons, neutrons and protons in a neutral atom of carbon-14 and how they are arranged.

symbols:

electron ×

neutron ○

proton ●

Fig. 8.1

[3]

(b) Describe how the composition of a neutral atom of carbon-14 is different from the composition of a neutral atom of nitrogen-14 ($^{14}_{7}N$).

.....

..... [2]

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(c) Carbon-14 decays by beta (β) emission.

(i) State the name of a particle that is identical to a beta-particle.

..... [1]

(ii) Describe the change that takes place in carbon-14 as a beta-particle is emitted.

.....

..... [1]

(d) The half-life of carbon-14 is 5700 years.

A very old object is made of wood. It contains 1.2×10^{11} atoms of carbon-14. When it was manufactured, it contained 9.6×10^{11} atoms of carbon-14.

Determine the time that has passed since it was manufactured.

time passed = ... [3]

[Total: 10]

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